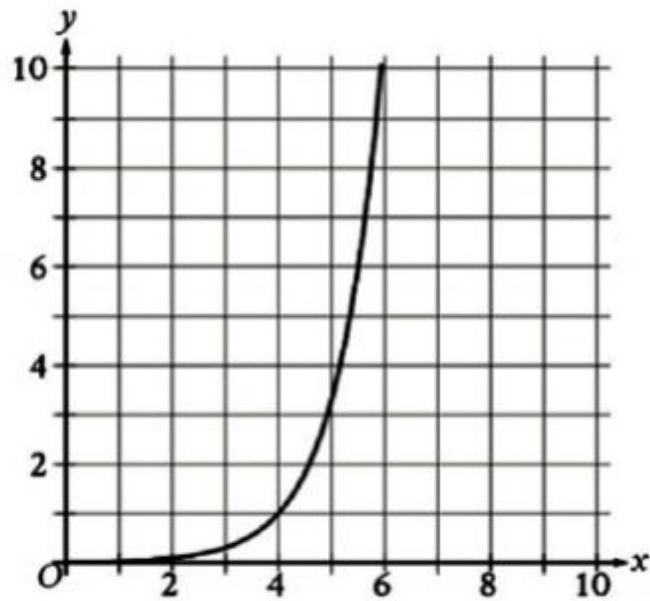


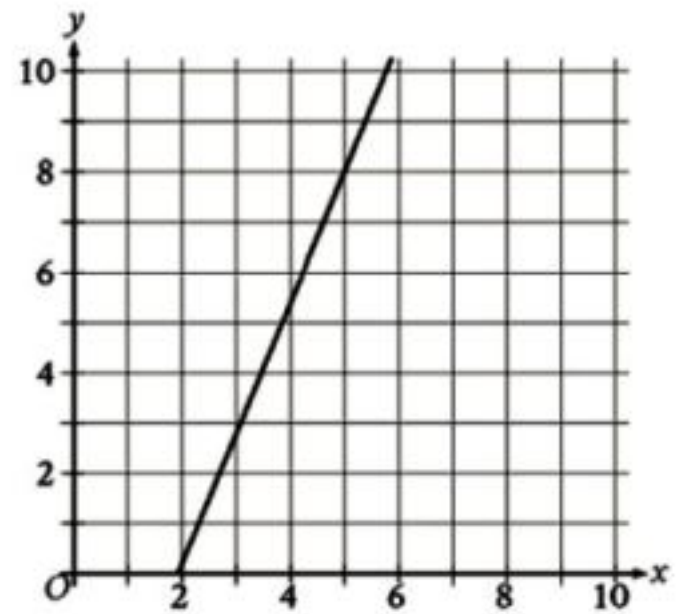
Math Mod2

1. Which of the following is a graph of a decreasing exponential function?

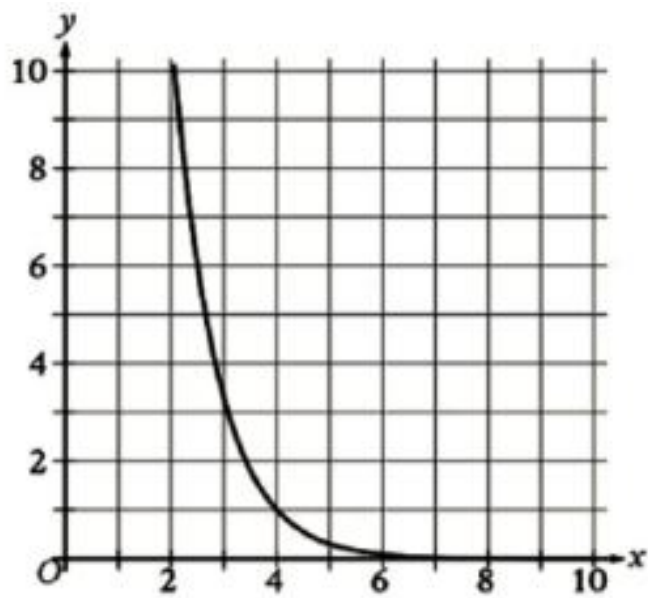
A)



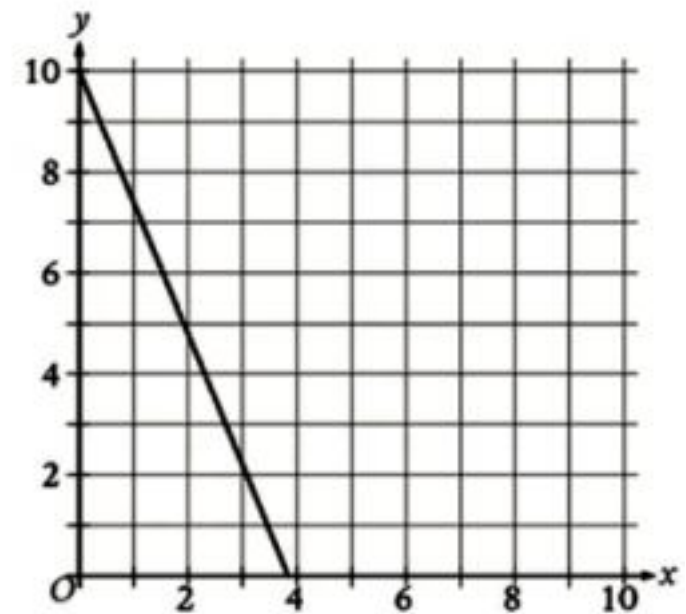
C)



B)



D)



2

What is the y-intercept of the graph of $y = 2x^2 + 6x + 3$ in the xy -plane?

- A) (0, 2)
- B) (0, 3)
- C) (0, 6)
- D) (0, 9)

3

The product of a positive number x and the number that is 1 less than x is equal to 342. What is the value of x ?

- A) 0.5
- B) 19
- C) 341
- D) 343

4

A cylinder has a diameter of 14 inches and a height of 8 inches. What is the volume, in cubic inches, of the cylinder?

- A) 49π
- B) 112π
- C) 392π
- D) $1,568\pi$

5

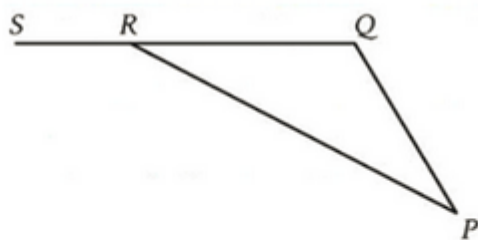
$$y = 2x^2 - 23x + 64$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A) -8
- B) -6
- C) 6
- D) 8

6



Note: Figure not drawn to scale.

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PQR$ is 142° , and the measure of $\angle PRS$ is 166° . What is the measure of $\angle QPR$?

- A) 14°
- B) 19°
- C) 24°
- D) 38°

7

Eli used wax to make candles. The function $f(x) = -15x + 60$ approximates the weight, in ounces, of wax Eli had remaining after making x candles. Which statement is the best interpretation of the y -intercept of the graph of $y = f(x)$ in the xy -plane in this context?

- A) Eli had approximately 15 ounces of wax when he began to make the candles.
- B) Eli used approximately 15 ounces of wax for each candle.
- C) Eli had approximately 60 ounces of wax when he began to make the candles.
- D) Eli used approximately 60 ounces of wax for each candle.

8

$$ak = \frac{b}{12}(7 + m)$$

The given equation relates the positive variables a , b , k , and m . Which equation correctly expresses m in terms of a , b , and k ?

- A. $m = \frac{12(ak-7)}{b}$
- B. $m = ak - 7 - \frac{b}{12}$
- C. $m = ak - \frac{7b}{12}$
- D. $m = \frac{12ak}{b} - 7$

9. The function g is defined by $g(x) = 10^{3x-1}$. What is the value of x when $g(x)$ is equal to 10,000?

- A) $\frac{4}{3}$
- B) $\frac{5}{3}$
- C) 3
- D) 4

10

If $\frac{x}{y} = 3$ and $\frac{5x}{ty} = 135$, what is the value of t ?

- A. $\frac{1}{9}$
- B. $\frac{3}{5}$
- C. $\frac{5}{3}$
- D. 9

11

A piece of string with a length of 35 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 5 more than 2 times the value of y . What is the value of x ?

12

Which equation has no solution?

- A. $-19x - \frac{1}{7} = -19x - \frac{1}{7}$
- B. $-19x - \frac{1}{7} = -19x + \frac{1}{7}$
- C. $-19x - \frac{1}{7} = 19x - \frac{1}{7}$
- D. $-19x - \frac{1}{7} = 19x + \frac{1}{7}$

13

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 8.80 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A) 15.84
- B) 47.84
- C) 475.83
- D) 507.83

14

An environmental scientist is investigating the volatility of 175 organic compounds by finding the boiling point of each compound. The mean boiling point of all 175 compounds is 167 degrees Celsius. The scientist classifies each of these compounds as either semi-volatile or volatile. Of the 175 compounds, 50 compounds were classified as semi-volatile, and these 50 compounds have a mean boiling point of 327 degrees Celsius. The remaining 125 compounds were classified as volatile. What is the mean boiling point, in degrees Celsius, of the 125 compounds classified as volatile compounds?

15

Line k is defined by $y = 6x + 5$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 11)$. Which equation defines line j ?

- A) $y = 6x + 11$
- B) $y = 11x + 5$
- C) $y = -6x + 11$
- D) $y = -11x + 5$

$$f(x) = 24(1.30)^{\frac{x}{3}}$$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

- A) 30
- B) 41
- C) 60
- D) 69

A carpenter charges a flat rate of \$234 for the first 3 hours of work and \$65 for each additional hour of work. Which equation gives the total amount y , in dollars, that the carpenter charges for w hours of work, where $w > 3$?

- A) $y = 65x + 39$
- B) $y = 234x + 65$
- C) $y = 65x + 234$
- D) $y = 234x + 429$

$$\sqrt[5]{119n}(\sqrt[6]{119n})^2$$

For what value of x is the given expression equivalent to $(119n)^{30x}$, where $n > 1$?

The table summarizes the distribution of age and assigned group for 105 participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	5	19	11	35
Group B	4	8	23	35
Group C	26	8	1	35
Total	35	35	35	105

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

- A) $\frac{2}{7}$
- B) $\frac{3}{7}$
- C) $\frac{19}{35}$
- D) $\frac{6}{7}$

x	$g(x)$
−20	4
−8	0
10	6

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+5}$ and f is a linear function. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. (0,-8)
- B. (0,5)
- C. (0,8)
- D. (0,40)

21

The mass of object A is 483% of the mass of object B, and the mass of object A is 0.069% of the mass of object C. If the mass of object C is p% of the mass of object B, what is the value of p/1,000 ?

22

$$\frac{1}{cx} = \frac{x}{152} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c?